
Can Generative Models Mimic Pokémon? Evaluating AI-Generated Creature Design in the Era of Copycat Games

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Abstract

Due to the recent advances in generative models, there have been questions about originality and intellectual property of certain products. In particular, the emergence of games such as Palworld have sparked debate over whether AI-generated content can mimic existing franchises like Pokémon. In this work, we investigate whether generative models can reproduce the visual style and design principles of Pokémon creatures. We train multiple generative models, which include GAN-based and diffusion-based models, to learn the underlying distribution of Pokémon images and generate new synthetic designs. We evaluate the generated images using Frechet Inception Distance (FID) and analysis of visual structure, stylistic consistency, and diversity. Our results show that simpler models achieve stronger performance and produce more coherent Pokémon-like images. Meanwhile, more complex models such as diffusion struggle in this low-data setting and generate less structured outputs. Overall, this study provides an analysis of generative mimicry and highlights the challenges of balancing similarity and originality in AI-generated design.

1 Introduction

In recent years, diffusion models, and generative adversarial networks (GANs) have become incredibly popular in the field of deep learning due to their remarkable image synthesis abilities. The fact that these models can learn about the complexities of various visual distributions, and still manage to produce very natural-looking and human-made images has opened doors to various creative fields in the art industry, such as game design and animation character design.

Within videogames, GANs are widely used particularly when it comes to creature design. With a long-standing franchise such as Pokémon, the genre has an idea of what constitutes a fair set of parameters. Games like Palworld have drawn a large amount of comparisons to the well-known style of Pokémon. Naturally, this raises questions about whether it's possible to train a GAN to mimic these styles.

Limitations of generative models in creative mimicry. Two key challenges arise when evaluating generative models in this context: (1) Can generative models capture stylistic patterns without directly copying existing designs? (2) How can we tell whether the generator is stylistically influenced by the training data or simply copying and pasting?

Human Classification of Generative Mimicry is distinct from other classification tasks in that it relies on highly subjective and often perceptually-driven criteria. Additionally, the fact that many current generative models are trained on a vast corpus of data sourced from the internet raises the question of how can a model avoid copying the copyrighted styles that it learns from this data in order to ensure that its output is truly original.

Motivation. This study explores the question of whether such generative models are suitable for emulating Pokémon style artwork; furthermore, the implications this has for future research and for companies working within the video game industry. It is an interesting exploration of the boundaries of such intellectual property and ownership of an artistic style.

Overview of the proposed approach. In this paper, we evaluate the ability of generative models to produce Pokémon-like creatures. We generate images using prompt-based diffusion models and analyze the outputs using both quantitative similarity metrics and qualitative evaluation. We compare generated images to official Pokémon designs to assess stylistic overlap and distinctiveness.

Our contributions are as follows:

- We provide a systematic evaluation of generative models in mimicking a defined visual style (Pokémon).
- We propose evaluation criteria for measuring stylistic similarity and originality in generated images.
- We analyze the implications of generative mimicry for creative industries and intellectual property.

2 Related Work

2.1 Generative Model Control and Steerability

This work explores how controllable generative models are with respect to desired outputs. Vafa et al. investigate the concept of *steerability*, which measures how effectively a model can be guided toward producing a target image. Their findings show that even state-of-the-art generative models often fail to reliably reproduce desired outputs, even with user-guided iterations only improving results in a limited number of cases.

This limitation is directly relevant to our setting. In generating Pokémon-like designs, the goal is not only to produce visually appealing images but also match a specific artistic style. Our results align with these findings, as models that produce lower FID scores still exhibit inconsistencies in structure and style, suggesting that generative mimicry is not purely a function of model quality but also of controllability.

2.2 Constraint-Aware Generative Modeling

This work focuses on enforcing constraints during sampling. Utkarsh et al. propose Physics-Constrained Flow Matching (PCFM). PCFM ensures that generated outputs satisfy strict physical constraints such as conservation laws. Unlike traditional approaches their method guarantees constraint satisfaction by modifying intermediate states during generation.

Although their work is focused on scientific applications, the idea of enforcing constraints is highly relevant to our task. Pokémon designs follow implicit structural and stylistic rules. Our findings suggest that generative models struggle to capture these constraints, particularly in diffusion-based approaches. This indicates that incorporating explicit constraints or domain-specific priors may improve generative performance in structured visual domains.

2.3 Alignment in Generative Models

Alignment is another critical challenge in generative modeling. Lu et al. demonstrate that 3D generative models often produce misaligned outputs due to inconsistencies in training data. To address this, they introduce orientation-aligned generation, which enforces consistent object orientation across categories and improves downstream usability.

This notion of alignment extends naturally to our problem. Pokémon designs exhibit consistent spatial structure and recognizable orientation. However, we observe that several generative models fail to maintain this consistency, leading to distorted or unnatural outputs

2.4 Summary

Overall, existing work highlights three key challenges in generative modeling: limited controllability; difficulty enforcing constraints; and inconsistencies in alignment. Our work builds on these insights by examining how these challenges manifest in a culturally significant and structured domain. Through our experiments, we show that these limitations persist even in modern generative models.

3 Methodology

3.1 Preliminaries

Problem definition. Assume we have a dataset of Pokémon images and related creature designs, where each image belongs to a distribution of stylized fantasy creatures. Let the input image space be denoted as $X \in \mathbb{R}^{H \times W \times C}$, where H and W are the image height and width, and C is the number of channels. Additionally, let $p \in \mathcal{P}$ denote a text prompt describing a creature (e.g., “electric mouse creature” or “small fire dragon”).

We consider a generative model G that maps a prompt p and latent noise z to an image x :

$$x = G(p, z), \quad z \sim \mathcal{N}(0, I).$$

Our goal is to evaluate whether G can generate images that mimic the Pokémon aesthetic while maintaining perceptual originality. In particular, we analyze whether prompt-conditioned generation can reproduce stylistic patterns such as color schemes, proportions, and facial structure that are characteristic of Pokémon designs.

Generation space. We define a set of prompts:

$$\mathcal{P} = \{p_1, p_2, \dots, p_K\},$$

where each prompt describes a Pokémon-like creature using attributes such as elemental type, animal features, or cartoon style.

For each prompt p , we generate multiple images by sampling different latent noise vectors z . This allows us to analyze both consistency and diversity in generated outputs. The prompt set is fixed across all experiments to ensure a controlled comparison.

Model space. In this work, we primarily focus on diffusion-based generative models, which have shown strong performance in text-to-image synthesis. Diffusion models generate images by iteratively denoising a random noise vector conditioned on a text prompt.

While other generative models such as GANs and VAEs exist, our experiments are based on a pretrained diffusion model due to its ability to produce high-quality and semantically aligned images.

Evaluation procedure. To evaluate a generated image $x = G(p, z)$, we define:

$$P(G, p) = \{\text{Sim}, \text{Style}, \text{Novelty}\},$$

where:

- Sim measures visual similarity to real Pokémon images,
- Style measures consistency with Pokémon design patterns,
- Novelty measures how distinct the generated image is from existing designs.

These metrics allow us to quantify the trade-off between mimicry and originality.

3.2 Overview

We propose a systematic framework for evaluating whether generative models can mimic Pokémon-style creatures. First, we define a standardized set of prompts representing Pokémon-like descriptions. Second, we generate multiple images per prompt using a diffusion-based model. Third, we evaluate generated outputs using similarity, stylistic consistency, and novelty metrics.

The framework is designed to isolate the effect of prompt-conditioned generation while keeping all other factors fixed. This allows us to analyze how well diffusion models capture stylistic features of Pokémon designs.

3.3 Generation Pipeline

For all experiments, we use a prompt-based diffusion generation pipeline:

- define a set of prompts describing Pokémon-like creatures;
- sample latent noise vectors $z \sim \mathcal{N}(0, I)$;
- generate images using a pretrained diffusion model $x = G(p, z)$;
- repeat generation multiple times per prompt to capture variation.

Diffusion-based generation. Diffusion models generate images through a denoising process that gradually transforms noise into structured images. Given a prompt p , the model learns to reverse a noise process and produce an image aligned with the prompt.

This approach is particularly effective for our task because it enables fine-grained control over semantic attributes while preserving global coherence in the generated images.

Prompt design. Prompts are carefully constructed to reflect common Pokémon characteristics, including:

- elemental attributes (e.g., fire, water, electric),
- animal-inspired features,
- cartoon-like visual style.

This ensures that generated images are semantically aligned with the Pokémon domain while allowing variation in design.

Each generated sample is then evaluated using the defined metrics to assess whether it resembles Pokémon-style creatures without directly replicating existing designs.

4 Experiments

4.1 Set Up

Dataset & Task. We evaluate generative models on the task of producing Pokémon-like creature designs. The objective is to learn the underlying distribution of Pokémon-style images and generate new samples that resemble the original data while maintaining diversity.

This task presents several challenges. First, Pokémon designs contain certain consistent visual patterns. Some of these include simplified shapes and distinct color schemes. Second, generative models may overfit and memorize training samples instead of producing novel outputs. Third, evaluating generation quality requires measuring both the similarity to real data and the diversity between generated samples.

Models. We train generative models that learn a mapping from a latent space to the image space. Given a latent vector $z \sim \mathcal{N}(0, I)$, the model generates an image:

$$x = G(z)$$

Evaluation Metrics. We evaluate generated images using:

- **Frechet Inception Distance (FID)** to measure distribution similarity,
- **Visual quality** (sharpness and realism),
- **Diversity** across generated samples.

Lower FID indicates better alignment with the real data distribution.

Table 1: Frechet Inception Distance (FID) for generative models on Pokémon-style image generation. Lower values indicate better performance.

Model	FID (↓)
DCGAN	206.7309
WGAN-GP	228.1761
Diffusion	354.5497
WAE	364.5446
VAE	411.1357
WINN	446.6009

4.2 Performance Evaluation

We report our results in Table 1. Our experiments demonstrate three key findings: (a) generative performance varies significantly across model types, (b) diffusion models do not necessarily outperform simpler architectures in this setting, and (c) dataset size and structure play a role in generative quality.

First, we observe that simpler generative models outperform more complex ones in generating Pokémon-like images. These models produce more recognizable structures tied to the Pokémon genre. These include coherent shapes, consistent coloring, and visually meaningful features. This suggests that they are better suited for learning from relatively small and structured datasets.

Second, diffusion models perform worse than expected. Historically, diffusion models have shown state-of-the-art results on large-scale natural image datasets. Despite this, they struggle in this setting where they generated images often appear blurry, lack clear structure, and fail to capture defining Pokémon characteristics. Due to this, the generated images appear similar to silhouettes; furthermore, this is reflected in higher FID scores, indicating poor alignment with the true data distribution.

Third, the worst-performing model (WINN) fails to generate meaningful images entirely; additionally, it has produced noisy images despite consistent fine-tuning. This suggests instability in training or an inability to capture the underlying data distribution.

Overall, these results indicate that model complexity does not guarantee better performance. In smaller, domain-specific datasets such as Pokémon, simpler models may generalize better; meanwhile, more complex models like diffusion require significantly more data to perform effectively.

5 Conclusion and Future Work

In this paper, we investigated the ability of generative models to mimic Pokémon-style image designs. Our experiments reveal several important findings regarding model performance, data limitations, and the trade-off between similarity and originality in generative modeling.

First, we observe that simpler generative models outperform more complex architectures in a smaller setting. In particular, DCGAN achieves the lowest Frechet Inception Distance (FID); due to this, it indicates that DCGAN has best alignment with the true data distribution. The generated images from DCGAN exhibit coherent structures, recognizable shapes, and visually consistent color patterns, suggesting that the model can effectively capture key characteristics of Pokémon designs.

Second, more advanced generative models such as diffusion models perform poorly than expected. Despite their success on large-scale natural image datasets, diffusion models have a poor performance on smaller datasets. They produced images that are often blurry, lack clear structure, and fail to capture defining features of Pokémon-style designs. This is further supported by higher FID scores compared to GAN-based approaches. These results suggest that diffusion models require substantially more data and training resources to perform effectively.

Third, the worst-performing model (WINN) fails to generate meaningful outputs, producing images that lack recognizable structure and consistency. This indicates instability in training or an inability to learn the underlying data distribution. Similarly, VAE-based approaches exhibit high FID scores, suggesting that they struggle to capture fine-grained visual details required for stylistically consistent image generation.

Overall, our findings highlight that model complexity does not guarantee better performance. Instead, the effectiveness of generative models depends heavily on dataset size, structure, and the ability of the model to capture relevant visual patterns. In smaller domain-specific datasets, simpler architectures may generalize better than more complex models.

For future work, several directions can be explored. First, increasing dataset size or incorporating data augmentation may improve the performance of more complex models such as diffusion. Second, hybrid approaches that combine GAN-based training with diffusion-based refinement could potentially balance image quality and diversity. Finally, incorporating domain-specific constraints or style losses may help guide generative models toward producing more realistic and novel Pokémon-like designs.

6 Bonus Points

This project explores a novel and relevant application of generative models by examining whether they can mimic Pokémon-style designs. This question is particularly timely given recent discussions around AI-generated content and games such as Palworld, where concerns about imitation and originality have become increasingly important.

Beyond standard implementation, this work provides a comprehensive comparison across multiple generative model architectures, including DCGAN, WGAN-GP, diffusion models, VAE, WAE, and WINN. We evaluate these models using both quantitative metrics such as Frechet Inception Distance (FID) and qualitative analysis of generated images. This allows us to provide deeper insight into how different architectures perform under the same dataset and task constraints.

Importantly, our findings challenge common assumptions in deep learning. While more advanced models such as diffusion are often expected to outperform simpler approaches, our results show that this is not always the case. Instead, performance depends heavily on dataset size and model compatibility with structured visual domains.

Overall, this project goes beyond standard coursework by combining a novel application, multiple model comparisons, and critical analysis of results. The insights gained from this work provide practical guidance for selecting generative models in real-world scenarios where data is limited and style consistency is important.

References

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